

SITE / FACILITY INFORMATION			
FACILITY:	Cadillac WWTP	Digestion Method: 1 (Anaerobic = 1, Aerobic = 2, Lime stabilized = 3) App. Method: Injection (Inj, Inc, Sfc, not case sensitive)	
MONTH & YEAR:	April 17, 2017		
FACILITY NPDES #	MI0020257		
MDEQ SITE ID #:	21N09W17-EK01		
COUNTY:	Wexford		
TOWNSHIP:	Clam Lake		
SECTION:	17		
OWNER:	Eugene Keelean	Org. N-1 yr. ago:	
FARMER:	John Rydquist	Org. N-2 yrs. ago:	
ADDRESS:	9897 S. 43 Rd.		
ADDRESS:	Cadillac, MI 49601	TOTAL ACRES:	40
TELEPHONE:	231-884-0719	ACRES USED:	23
SUPERINTENDANT	Jeff Dietlin		
		App. Rate (Gal/Ac)	18,221
LATITUDE:	N44° 20'86"	App. Rate (DT/Ac)	1.94
LONGITUDE:	W85° 43'58"	Org. N App. (#/ac)	206

SOIL TEST RESULTS			
SAMPLE DATE:	07/27/16	NEXT CROP:	Corn
CEC:	7.6	CROP YLD GOAL:	160
PH:	6.1	NIT. REC. (lbs/ac):	190
LIME INDEX:	6.9	ADJ. N REC.	190
BRAY P1 (ppm):	100	AVAIL. N APP.	189
POTASSIUM (ppm):	80	SOIL & APP. P	357
CALCIUM (ppm):	1102	P ₂ O ₅ REC. (lbs/ac):	0
MAGNESIUM (ppm):	77	K ₂ O REC. (lbs/ac):	70
		LIME REC. (T/ac):	1 ta

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ZINC (Zn)	9.53	lbs/acre
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	BIOSOLIDS ANALYSIS	
Lab ID	KAR	703913
Sample Name		Cadillac WWTP
Date Collected		3/28/2017
Matrix		Sludge
Density	lbs/gal	8.45
Mercury	mg/kg	0.35
Nitrogen,Ammonia	mg/kg	38,100.00
Nitrogen,Total	mg/kg	0.00
Nitrogen,Total Avail.	lb/ton	0.00
Nit.,Total Kjeldahl	mg/kg	91,200.00
pH	STD Units	8.10
Total Phosphorus	mg/kg	40,400.00
Nitrogen,Nitrate	mg/kg	20.00
Cadmium	mg/kg	2.50
Calcium	mg/kg	29,300.00
Chromium	mg/kg	97.20
Copper	mg/kg	1,020.00
Lead	mg/kg	31.40
Magnesium	mg/kg	5,220.00
Molybdenum	mg/kg	25.00
Nickel	mg/kg	29.30
Potassium	mg/kg	4,170.00
Zinc	mg/kg	1,480.00
Arsenic	mg/kg	26.60
Selenium	mg/kg	6.30
Total Solids	%	2.52
Total Volatile Solids	%	57.20

BIOSOLIDS APPLIED

[illegible]

Source:	Cadillac WWTP	Owner:	Eugene Keelean	App. Rate (Gal/Ac)	18,221
Date:	04/17/17	Farmer:	John Rydquist	App. Rate (DT/Ac)	1.94
NPDES #:	MI0020257	Address:	9897 S. 43 Rd.	Useable Acres	40.0
MDEQ Site #:	21N09W17-EK01	City:	Cadillac, MI 49601	Acres Applied	23.0
County:	Wexford	Telephone:	231-884-0719	Seasons Used	4
Township:	Clam Lake	Latitude:	N44° 20'86"		
Section:	17	Longitude:	W85° 43'58"		

BIOSOLIDS ADDITIONS									
<u>Nutrient</u>	<u>Biosolids Additions</u>		<u>Residual Organic N Credit</u>		<u>Soil Fertility Test</u>			<u>Total Estimated Nutrients</u>	
Nitrogen (lb N/Ac)	189	+	0	+	0	lb N	=	189	lb N/Ac
Phos (lb P/Ac)	157			+	200	lb P	=	357	lb P/Ac
Pot (lb K/Ac)	16			+	160	lb K	=	176	lb K/Ac

SOIL ANALYSIS AND CROP INFORMATION					
C.E.C.:	7.6	P (lbs/acre):	200	K (lbs/acre):	160
Soil pH:	6.1	P (ppm):	100	K (ppm):	80
Lime Index:	6.9	Ca (lbs/acre):	2204	Mg (lbs/acre):	154

FERTILIZER RECOMMENDATIONS (lbs./a)						
Crop to be Fertilized	Yield Goal	N	P2O5	K2O	Lime	
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Corn	160	190	0	70	1 ta	Lbs./A

BIOSOLIDS ANALYSES									
Date: 28-Mar		Dry wt. basis	Lbs/dry ton	Lbs Applied Per Acre	Allowable Lifetime Lbs/acre	Previous Applied lb/acre	Total Applied lb/acre	Life in Years	% Cum.
		-----	-----	-----	-----	-----	-----	-----	-----
Density	(mg/l)-->	1.01	----	----					
Weight	(Lb/Gal)>	8.45	----	----					
Solids	(%)----->	2.52	----	----					
TKN	(%)----->	9.12	----	----					
Amm. N	(%)----->	3.81	----	----					
Nit. N	(%)----->	0.002	----	----					
Total Plant Avail. N----->			97	189					
Total P	(%)--->	4.04	81	157					
Total K	(%)--->	0.42	8	16					
Total Ca	(%)--->	2.93	58.60	113.68					
Total Mg	(%)--->	0.52	10.44	20.25					
Total As	(ppm)->	26.60	0.053	0.103	37	0.15	0.25	359	0.68%
Total Cd	(ppm)->	2.50	0.005	0.010	35	0.04	0.05	3,608	0.14%
Total Cr	(ppm)->	97.20	0.194	0.377	N/A	0.68	1.06	N/A	N/A
Total Cu	(ppm)->	1020.00	2.040	3.958	1,335	6.17	10.13	337	0.76%
Total Pb	(ppm)->	31.40	0.063	0.122	267	0.31	0.43	2,192	0.16%
Total Hg	(ppm)->	0.35	0.001	0.001	15	0.00	0.00	11,046	0.01%
Total Mo	(ppm)->	25.00	0.050	0.097	N/A	0.17	0.27	N/A	N/A
Total Ni	(ppm)->	29.30	0.059	0.114	274	0.19	0.30	2,410	0.11%
Total Se	(ppm)->	6.30	0.013	0.024	89	0.05	0.07	3,641	0.08%
Total Zn	(ppm)->	1480.00	2.960	5.742	2,492	9.53	15.27	434	0.61%

42842

Facility NPDES # MI0020257

Lat/Long: N44* 20'86"

Total acres in site: 40

fertilized: **Corn**

pH 6.1 S.U.

K 160 lbs/a

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4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
4.04		0.42		26.60		2.50		97.20		1,020.00		31.40		0.35		25.00	
157		16.18		0.103		0.010		0.377		3.958		0.122		0.001		0.097	
157		16.18		0.103		0.010		0.377		3.958		0.122		0.001		0.097	
176		18.12		0.116		0.011		0.422		4.432		0.136		0.002		0.109	
-----				0.253		0.050		1.057		10.128		0.432		0.001		0.267	
-----				0.284		0.056		1.184		11.343		0.484		0.002		0.299	

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3605.22	372.123	2.37373	0.223095	8.673938	91.0228	2.802075	0.031233	2.230951
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As	37
Cd	4.5
Cu	190
Pb	268
Hg	15
Ni	76
Se	89
Zn	380

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29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
29.30		6.30		1,480.00		2.93		0.52		Err:509
0.114		0.024		5.742		113.682		20.253		Err:509
0.114		0.024		5.742		113.682		20.253		Err:509
0.127		0.027		6.431		127.323		22.684		Err:509
0.304		0.074		15.272		113.682		20.253		Err:509
0.340		0.083		17.105		127.323		22.684		Err:509

0.19	0.05	9.53	0	0	0
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Application Date	Facility	Facility #	Site I D	County	Township	Section
April 17, 2017	Cadillac WWTP	MI0020257	21N09W17-EK01	Wexford	Clam Lake	17

Owner	Farmer	Farmer Street Ad	Farm City, St.	Zip	Frm Telephone	Entity Super	Latitutde
Eugene Keelean	John Rydquist	9897 S. 43 Rd.	Cadillac, MI	49601	231-884-0719	Jeff Dietlin	N44* 20'86"

Longitude	Season Used	Dig Method	App Method	Ttl Acres	Acres Used	Soil Date
W85* 43'58"	4	1	Injection	40	23	7/27/2016

CEC	Ph	Lime Index	Bray P1	Pot	Calcium	Magnesium
7.6	6.1	6.9	100	80	1102	77

Intend Crop	Yld Goal	N Rec	Phos Rec	Pot Rec	Lime Rec
Corn	160	190	0	70	1 ta

Facility	Facility #	Lab I.D.	Lab Ref #	Sample ID	Date Collected	Matrix	Density
Cadillac WWT	MI0020257	KAR	703913	Cadillac WW	3/28/2017	Sludge	8.45

Hg	N Amonia	N Total	N Ttl.AvaIL #/T	N Ttl Kjeldahl	pH	Ttl. Phos	N Nitrate
0.35	38100	0	0	91200	8.1	40400	20

Cd	Ca	Cr	Cu	Pb	Mag	Mo	Ni
2.5	29300	97.2	1020	31.4	5220	25	29.3

K	Zn	As	Se	Ttl. Solids	Ttl. Vol. Solids
4170	1480	26.6	6.3	2.52	57.2

Facility	Facility #	Site ID	Date 1	Gallons 1	% Solids 1	Date 2	Gallons 2
adillac WW	MI0020257	1N09W17-EK	4/17/2017	71,040	2.52	4/18/2017	70,980

% Solids 2	Date 3	Gallons 3	% Solids 3	Date 4	Gallons 4	% Solids 4	Date 5
2.52	4/25/2017	120,768	2.52	4/27/2017	156,288	2.52	

Gallons 5	% Solids 5	Date 6	Gallons 6	% Solids 6	Date 7	Gallons 7	% Solids 7
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Date 8	Gallons 8	% Solids 8	Date 9	Gallons 9	% Solids 9	Date 10	Gallons 10
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% Solids 10	Date 11	Gallons 11	% Solids 11	Date 12	Gallons 12	% Solids 12	Date 13
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Gallons 13	% Solids 13	Date 14	Gallons 14	% Solids 14	Date 15	Gallons 15	% Solids 15
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Date 16	Gallons 16	% Solids 16	Date 17	Gallons 17	% Solids 17	Date 18	Gallons 18
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% Solids 18	Date 19	Gallons 19	% Solids 19	Date 20	Gallons 20	% Solids 20	Date 21
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Gallons 21	% Solids 21	Date 22	Gallons 22	% Solids 22	Date 23	Gallons 23	% Solids 23
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Date 24	Gallons 24	% Solids 24	Date 25	Gallons 25	% Solids 25	Date 26	Gallons 26
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% Solids 26	Date 27	Gallons 27	% Solids 27	Date 28	Gallons 28	% Solids 28	Date 29
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Gallons 29	% Solids 29	Date 30	Gallons 30	% Solids 30	Date 31	Gallons 31	% Solids 31
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